but general relativity shows that it fails. The more gas you pump in, the more massive the sphere's contents become. And before you reach the entropy of an equal-sized black hole, the increasingly large mass within the sphere will reach a critical value that causes the sphere and its contents to *become a black hole.* There's just no way around it. Black holes have a monopoly on maximal disorder.

What if you try to further increase the entropy in the space inside the black hole itself by continuing to pump in yet more gas? Entropy will indeed continue to rise, but you'll have changed the rules of the game. As matter takes the plunge across a black hole's ravenous event horizon, not only does the black hole's entropy increase, but *its size increases as well.* The size of a black hole is proportional to its mass, so as you dump more matter into the hole, it gets heavier and bigger. Thus, once you max out the entropy in a region of space by creating a black hole, any attempt to further increase the entropy in that region will fail. The region just can't support more disorder. It's entropy-sated. Whatever you do, whether you pump in gas or toss in a Hummer, you will necessarily cause the black hole to grow and hence surround a larger spatial region. Thus, the amount of entropy contained within a black hole not only tells us a fundamental feature of the black hole, it also tells us something fundamental about space itself: *the maximum entropy that can be crammed into a region of space—any region of space, anywhere, anytime—is equal to the entropy contained within a black hole whose size equals that of the region in question.*

So, how much entropy does a black hole of a given size contain? Here is where things get interesting. Reasoning intuitively, start with something more easily visualized, like air in a Tupperware container. If you were to join together two such containers, doubling the total volume and number of air molecules, you might guess that you'd double the entropy. Detailed calculations confirm[1] this conclusion and show that, all else being equal (unchanging temperature, density, and so on), the entropies of familiar physical systems are proportional to their volumes. A natural next guess is that the same conclusion would also apply to less familiar things, like black holes, leading us to expect that a black hole's entropy is also proportional to its volume.